



Effect of the PTO Damping Force on the 2D Oscillating Buoy Wave Energy Converter Integrated into a Pile-Restrained Floating Breakwater

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Abstract

The aim of the study is to evaluate the parametric effect of a power take-off (PTO) system on an integration system of an oscillating buoy wave energy device with a pile-restrained floating breakwater. To consider the viscous effect, the study is carried out with three conditions using OpenFOAM, i.e., the linear optimal PTO system, the fixed PTO damping coefficient, and the Coulomb PTO system. The results show that the maximum wave energy capture ratio (CWR) of the device can merely reach 0.35 in the present work, when considering the viscous effect. This result is smaller than that obtained from potential flow. In addition, the linear optimal PTO coefficient under viscous flow should be larger than the theoretical optimal coefficient in terms of the potential flow. A comparison of the Coulomb damping model and the linear damping model also shows that the Coulomb damping model has a good performance in terms of both wave energy capture efficiency and coastal protection. Hence, the effect of the PTO system on such an integration system can be an essential factor to achieve an optimal design.

Keywords Wave energy converter (WEC) · Integration system · Power take-off (PTO) · OpenFOAM

1 Introduction

With the consumption of fossil fuels around the world, the climate has changed rapidly, with serious environmental issues. Humankind has to seek cleaner sources of energy for sustainable development. As a clean energy source, renewable energy sources such as wind energy, solar energy, and ocean energy have received widespread attention and have been under development for a long time. Among these kinds of renewable energies, ocean energy, especially ocean wave energy, has received wide research and development. Falcao (2010) provided a detailed overview of the current wave energy utilization and research; clear classification of the wave energy converter system (WEC) has been carried out.

The WEC devices that have been tested and prototyped at present are introduced in detail. At the same time, the power take-off system (PTO) and the mooring system are also classified and are given a detailed introduction.

However, the high construction costs and the low energy conversion ratio of the WEC are still the reasons that limit the widespread use and development of the WEC. On the one hand, the cost can be reduced by increasing the wave energy conversion rate. Several WEC devices with efficient energy conversions have been designed and studied, and some have taken the prototype test. On the other hand, recently, there have been some attempts to reduce costs through cost sharing by integrating WEC devices with existing marine structures such as the integration of a WEC with a breakwater. Such an integrated system can reduce the construction costs by the cost sharing concept. At the same time, this system can enhance the stability and extend the operating life of the WEC device.

For the integration of a WEC and breakwater, Mustapa et al. (2017) introduced the integration system of an oscillating water column (OWC) and shore-based breakwaters in their overview paper and summarized the advantages and problems of integrated systems in the current stage. At the same time, the paper also proposed that new integration

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concepts of wave energy device and breakwater structure need to be developed to achieve high performance. However, the paper did not talk about the integration of floating structures and WEC systems, which have a considerable potential to be developed. Because the floating breakwater requires a lower cost, especially under deep water conditions, and because at the same time it is beneficial to the exchange of water and has a lower environmental impact, the floating breakwater has been widely used and studied. The research on floating breakwaters has made many achievements that draw the floating breakwater into engineering practice.

The crucial problems with the integration of a WEC and the floating breakwater are the optimizing of the wave energy converter efficiency and the wave damping performance. In this regard, Zanuttigh et al. (2010) pioneered the concept of applying the WEC device to coastal defences. An initiative designed WEC device named DEXA which was used in the experimental study. The effect of characteristic length and different wave steepness on the wave capture performance and coastal defence are evaluated carefully. Further optimization needs to be performed to obtain better performance. He and Huang (2014) conducted a physical experimental study on a pile-supported OWC device as a breakwater. The results show that such structures have the function of a breakwater. Meanwhile, the structure can take into account the wave power generation. Based on the principle of an oscillating buoy WEC, Zhao et al. (2017) carried out a theoretical analysis of the WEC integrated with a floating breakwater under the assumption of linear and inviscid flow. The linear PTO system is applied to all cases. In addition, they found that this integration system can be used as a WEC device and that it reduces the transmissive waves in a specific range of wave conditions.

However, the design of a WEC device often makes the device resonate under the action of waves. Under these circumstances, the resonant motion is very sensitive to the viscous effect. At the same time, the viscous and nonlinear effects also dissipate the energy, thus reducing the energy conversion efficiency. Furthermore, the device is unable to solve a strong nonlinear wave and extreme sea state problems under the linear and inviscid hypotheses. Therefore, most of the research on such problems is mainly based on experiments or numerical simulations based on viscous flow.

The power take-off (PTO) system is an important part of the integration system accounting for the energy conversion efficiency. This system also provides extra restraint force to the buoy that will influence the response of the floating buoy. With the changes in the dynamic response, the energy conversion efficiency and the wave damping performance will also be affected. Thus, it is necessary to study the performance of the PTO system in such an integration system.

Zhao et al. (2017) provided the theoretical analysis of the WEC integrated with a floating breakwater with a linear

PTO system. The energy conversion efficiency and the wave damping performance of the integration system are studied with different times of the optimal PTO coefficient λ_{optimal} . The result shows that a broader effective wave range can be obtained with $1.5 - 2.0\lambda_{\text{optimal}}$ if taking the transmission coefficient $K_t < 0.5$ and can capture a width ratio $\text{CWR} > 0.2$ as the criterion. However, as mentioned in the paper, the result is based on the assumption of linear and inviscid flow. The result will lead to a dramatic error at the resonance interval. In addition, further research is needed to determine the optimal PTO coefficient under viscous flow. The Coulomb damping model can be a simple model of a hydraulic PTO system, which is commonly used in many energy conversion devices. Ning et al. (2016) applied the Coulomb damping PTO system to such an integration system as the model in Zhao's work. In addition, an experiment was performed to research the performance of such an integration system with different PTO damping forces. The system can also achieve good performance with a proper adjustment of the PTO damping coefficient and the structure parameters. The experiment lacks a comparison of the linear PTO and the Coulomb PTO, which may still have some disputes about the performance of these two kinds of PTO system. Thus, to consider the energy conversion efficiency and the wave damping performance at the same time, further verification and research work should be conducted on the effect of a PTO system on the integration system.

Consider that the integration system is an integration of a WEC and pile-restrained floating breakwater, which can be approximated as a 2D dimension structure. The problem under consideration corresponds to normally incident waves interacting with an infinitely long horizontal cylinder restrained by piles. The 3D effect to the diffraction and radiation could be neglected, and also the wave conditions in the study are still mild without overlapping and wave breaking. A 2D section of the system is simulated as in Zhao's (2017) analytical work or Ning's (2016) experiment set-up. After building the numerical wave tank using the open-source CFD software OpenFOAM (2011), an integrated system of a two-dimensional WEC and breakwater is simulated. The current study aims at revealing the effect of the PTO damping system on the integration system. The linear optimal PTO damping characteristics and the hydrodynamic performance of the WEC-type breakwater integrated system under optimal damping are studied. Instead of turning the PTO damping corresponding to the wave conditions, fixed damping coefficient PTO systems keep various wave conditions constant. A comparison is given in terms of the capture width ratio (CWR) and the wave damping performance between the optimal PTO damping forces and the fixed PTO damping forces. In addition, based on the content of the Coulomb damping systems mentioned above, a Coulomb damping PTO system model is developed in OpenFOAM.

Then, we study the hydrodynamic performance of the WEC-type breakwater integration system with the Coulomb PTO damping and its difference from the linear PTO system.

2 Materials and Methods

The WEC-type breakwater integrated system is calculated using the interDyMFOam solver in OpenFOAM for the hydrodynamics. Using the single fluid (mixture) assumption, the incompressible two-phase Navier–Stokes equations are solved by the finite-volume method. The continuity Eq. (1) and momentum Eq. (2) can be written as:

$$\nabla \cdot u = 0 \quad (1)$$

$$\frac{\partial}{\partial t}(\rho u) + \nabla \cdot (\rho(u - u_g)u) = -\nabla p + \nabla \cdot S + \rho f_b + s\rho u \quad (2)$$

where u is the velocity of the fluid, u_g is the mesh velocity when updating the mesh (Ferziger and Peric 2012), ρ is the mixture density for two-phase flow, p is the total pressure, S represents the viscous stress tensor which can be expressed as $S = 2\mu D$, μ is the viscosity of the mixture, D is the rate-of-strain tensor, the gravitational force and the surface tension is denoted by f_b , and $s\rho u$ is the source term that can be used to set the damping zone for wave damping.

The VOF method (Hirt and Nichols 1981) is used to treat the interface of the two phases. The principle of the VOF method is to define the volume fraction α , whose value varies in the range [0, 1]. The value 0 represents the air phase, and the value 1 represents the water phase. The value at the interface of the two phases varied between 0 and 1. Therefore, it can track the free surface. Meanwhile, the density and the viscosity of the mixture can be donated by a weighted average of the air and water using α as the weight. The forum can be presented as:

$$\rho = \alpha\rho_w + (1 + \alpha)\rho_a \quad (3)$$

$$\mu = \alpha\mu_w + (1 + \alpha)\mu_a \quad (4)$$

where ρ_w represents the density of the water phase, ρ_a represents the density of the air phase, and α represents the volume fraction. Then, the volume fraction of the fluid α can be updated by solving the transport equation, as follows:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha(u - u_g)) = 0 \quad (4)$$

A treatment of interface compression present in OpenFOAM is named MULES. This treatment can keep the sharp interface and avoid the serious dissipation. Therefore, it can track the free surface more accurately (Berberovic et al. 2009).

Waves are generated by a piston wavemaker, which extends from the external interface for a wavemaker in the olaFlow toolbox (Higuera et al. 2013). A piston wavemaker can avoid the surface increasing during the simulation using the velocity inlet method. In addition, a high calculation efficiency can be expected. The transfer function S can represent the relation between the motion of the paddle and the target wave. The relation is shown as:

$$\frac{H}{S_0} = \frac{4 \sin h^2(kd)}{2kd + \sin h(2kd)} \quad (5)$$

where S_0 represents the stroke of the paddle, H is the target wave height, k represents the wave number, and d is the water depth.

From the transfer function, we can obtain the expression for the displacement and velocity:

$$X(t) = \frac{S_0}{2} \sin(\omega t) \quad (6)$$

$$U(t) = \frac{S_0 \omega}{2} \cos(\omega t); \quad (7)$$

thus, the linear airy wave surface is:

$$\eta(x, t) = \frac{H}{2} \cos(kx - \omega t) \quad (8)$$

where $X(t)$ represents the displacement of the paddle, $U(t)$ represents the velocity of the paddle, t is the time, x the position of the paddle, and ω is the wave circular frequency.

A damping zone is implemented at the end of the numerical wave tank. This method was achieved by adding a source term $s\rho u$ to the momentum equation and by setting the effective range of the source term as the damping zone. Outside the damping zone, the source term becomes zero and has no effect on the solution of the common momentum equation (Cha 2011):

$$s = \begin{cases} \frac{x-x_0}{x_1-x_0} S_0, & x_0 < x < x_1 \\ 0, & 0 < x < x_0 \end{cases} \quad (9)$$

where x is the x coordinate in an arbitrary position of the wave tank, x_0 represents the starting point of the damping zone, and x_1 represents the end point of the damping zone. The damping factor is chosen from the work of Schmitt and Elsaesser (2015). A reasonable value of the damping factor between 0 and 5 can effectively prevent the reflection wave. After some test cases, the damping factor is set to 5 for all simulation cases. This factor shows that the simulation can achieve a stable and accurate wave with the damping factor. Meanwhile, wavemaker provides an active wave absorption method in the olaFlow, which can absorb the reflection wave

from the structure and the reflection wave from the damping zone so that the simulation can remain stable for a long time.

The rigid body dynamics of the WEC-type breakwater are solved using the rigid body motion solver named *sixdofRigidBodyMotionSolver* in OpenFOAM for solving motions in six degrees of freedom. The fluid force F_p and torque M_p acting on the body are obtained by integrating the fluid pressure p over its entire surface area S :

$$F_p = \iint_S p \hat{n} dS \quad (10)$$

$$M_p = \iint_S r_{CS} \times p \hat{n} dS \quad (11)$$

where r_{CS} represents the position vector from the centre of mass to each point at the structural surface. The integration of the pressure is in the global coordinate system. However, there are also external restraining force F_R and moment M_R acting on the body in the local body frame. Therefore, to get the total force F_{tot} and total torque M_{tot} acting on the body, the moment needs to be transformed to the local body frame before it is added to the total torque together with the contribution from the restraints F_R . The equation expression can be expressed as:

$$F_{tot} = F_p + \sum_j F_R^{(j)} \quad (12)$$

$$M_{tot} = Q \left(M_p + \sum_j r_{CR}^{(j)} \times F_R^{(j)} + M_R^{(j)} \right) \quad (13)$$

where F_R is the external restraining force, F_R is the external restraining moment, R_{cr} is the global frame position vector of the restraint point of action on the body relative to the centre of gravity, and Q represents the rotation matrix from the global to the local frame of reference, which can transform the moment in the global frame to the local frame of reference.

The PTO system implemented to the WEC-type breakwater includes the linear PTO restraint and the Coulomb damping type. The linear PTO restraint is implemented as:

$$F_{pto} = B_{pto} V_i \quad (14)$$

where B_{pto} is the damping coefficient that derives from the analytic solution based on the potential theory. The optimal PTO damping can be defined as:

$$\lambda_{optimal} = \sqrt{(K/\omega - \omega(M + \mu))^2 + \lambda^2} \quad (15)$$

where M represents the mass matrix, μ is the additional mass coefficient, λ corresponds to the radiation damping coefficient, and K is the stiffness for the structure in static water.

The Coulomb-type PTO is implemented as:

$$F_{pto} = C_{pto} \text{sign}(\dot{X}) \quad (16)$$

which has an infinite gradient when the velocity is zero. Hence, this type of Coulomb PTO is easy to achieve non-physical oscillation. An approximate model is often used in the simulation that is expressed as:

$$F_{pto} = \min(G_{pto} |\dot{X}|, C_{pto}) \text{sign}(\dot{X}) \quad (17)$$

where F_{pto} can become zero by still being a very large gradient when the velocity tends to be zero. It is easier to get a smooth numerical solution. These three kinds of PTO models (Babarit et al. 2012) described above are shown in Fig. 1.

3 Case set-up and Validation

To compare the experimental data, the verification model in this section is referenced to the work of Ning et al. (2016). A 2D pile-restrained floating breakwater without any extra restraint is calculated as the verification model. Using the Froude scaling, a 1:10 model was adopted. The dimensions of the floating breakwater model are a length of 0.8 m, a height of 0.6 m, and a draft $d=0.2$ m.

3.1 Numerical Wave Tank Validation

A high-quality 2D numerical wave tank was modelled. Compared with the method that inputs the velocity at the inlet according to the specific wave theory, the piston wavemaker

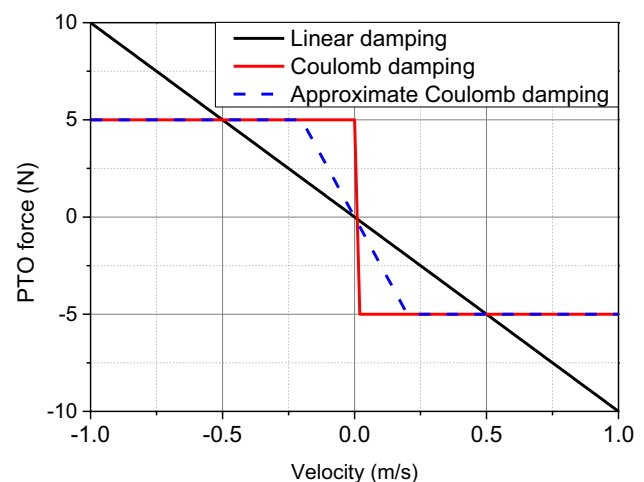


Fig. 1 Relationship between PTO force and velocity using three different damping models. (Babarit et al. 2012)

method in the olaFlow toolbox was chosen to generate regular waves. The length of the wave tank is 25 m, the height is 1.8 m, and the water depth is set to 1.0 m. The uniform polyhedral meshes were created using the blockMesh grid generation utility in OpenFOAM. The grid resolution was finer near the free surface because the VOF method needs more meshes to capture the free surface according to some previous works Davidson et al. (2015) and Kamath et al. (2015). Therefore, there were 10–20 meshes among the wave height and 60–120 meshes per wavelength. To make sure the numerical wave damping is small enough to be neglected along the tank, the ratio of the mesh length–width was under 2. The length of the damping zone is 1.5 times that of the wavelength. The damping coefficient s is 5. The left boundary is set as the inlet boundary. The velocity boundary condition U is *movingWall-Velocity*. The pressure boundary condition p_rgh is the *fixed-FluxPressure*, and the wavemakerMovement condition also needs to be defined in the *pointDisplacement* dictionary. Other boundary condition settings reference Higuera's work (2013) about the validation of the olaFlow toolbox. The adjusted time step method is set to control the time step, and the maximum Courant number is set to 0.5 to ensure numerical stability. To compare the physical experiment, the wave conditions are also referenced from the Ning's work. Considering the strong motion at the resonance wave period, a smaller interval is chosen near the resonance wave period. The wave conditions are shown in Table 1. The wave period T varied from 1.275 s to 2.0 s. The target incident wave height $H_i = 0.2$ m. The Re number is estimated to be 8×10^4 with an estimated average wave velocity of 0.1 m/s and the character length of 0.8 m. The KC number is less than 1.0. The turbulence effects in the laboratory scale might not be of importance. The laminar flow model is applied in the simulation, and the result shows good agreement with the theoretical wave surface elevation.

The calculation time is 35 s. A series of regular waves are generated by the piston wavemaker. The test wave conditions are $T = 1.58$ s and $H = 0.2$ m. Based on the wave theory, this condition is in the zone of the third-order Stokes wave theory. A probe is located in the point that is 2 m away from the inlet boundary. The surface elevation is in good agreement with the theoretical surface elevation under the third-order Stokes wave theory. The wave crest gets steep, the wave trough gets flat, and a rise in the mean sea level is also observed. From 15 s to 35 s, there are more than 10 complete wave surface elevations, as shown in Fig. 2. The error of the wave height is less than 1%. Thus, a high-quality numerical wave tank is validated.

Table 1 Test wave conditions for the laboratory model

d (m)	T (s)	Kh	H_i (m)
0.2	1.275, 1.37, 1.445, 1.488, 1.58, 1.653, 1.79, 2.0	2.513–1.178	0.2

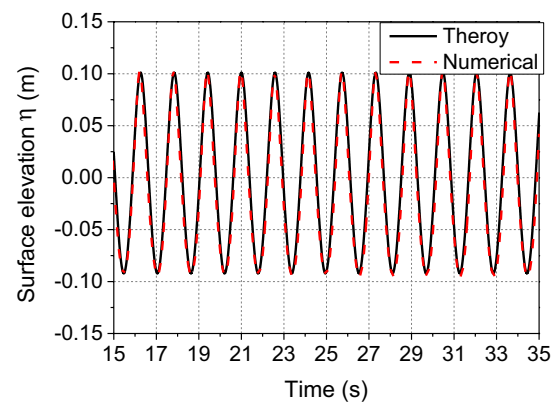


Fig. 2 Comparison of the time history of surface elevation between numerical model and theory model

3.2 Verification case for the simulation of floating structures

After the numerical wave tank validation, the pile-restrained floating breakwater is calculated as the verification in the above numerical wave tank. The floating breakwater was restrained by a pile at the test location that was 12 m away from the wavemaker that can ensure there is enough distance to both the wavemaker boundary and the damping zone. Meanwhile, to measure the surface elevations, there were four wave gauges installed in the wave tank in total. Two wave gauges were placed in front of the model for separating the incident wave height H_i and the reflection wave height H_r at the positions of 9.1 m and 9.6 m away from the wavemaker, respectively. In addition, the other two were placed on the leeward side of the model for getting an average transmission wave height H_t located at 14.1 m and 14.6 m away from the breakwater. The grid is generated by removing the meshes of the floating breakwater from the background mesh of the wave tank. To ensure the accuracy of the solution, the meshes were refined around the breakwater boundary, and the mesh convergence test was performed, as shown below. Figure 3 shows a sketch of the case set-up. As said in Ning's work "Note that for the pile, the dimensions of the cross section, whose size is $0.04 \text{ m} \times 0.03 \text{ m}$, are sufficiently small so that the influence of the slide rail on wave field can be neglected". The dash line presents the pile which is modelled as a linear or Coulomb damping force instead of a physical one. Figure 4 shows the grid resolution around the floating breakwater.

3.2.1 Mesh convergence test

To test the mesh convergence, the wave conditions are $T = 1.58$ s and $H = 0.2$ m when the floating breakwater resonates with the wave. Three groups of mesh resolutions have

Fig. 3 Sketch of the model and the wave tank set-up. WG: wave gauge, the pile in dash line is modelled as a constraint force instead of a physical one

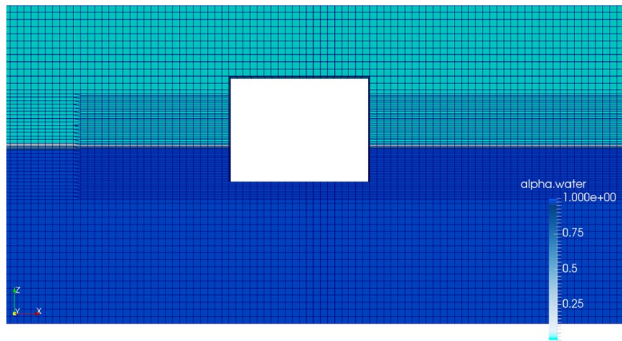
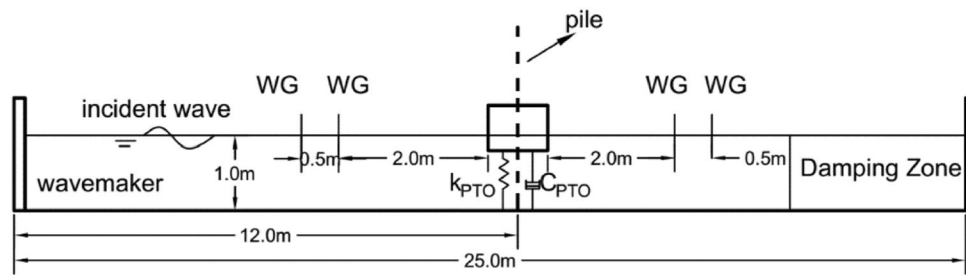


Fig. 4 Mesh configuration of the computational domain near the floating buoy; the model is built in x - z coordinate plane with one mesh in y -axis as shown in the figure

different numbers of meshes. In the coarse mesh group, the mesh is not refined around the breakwater boundary. Some tests show that the solution is not very sensitive to the size of the background mesh, except for a refined mesh near the free surface. Thus, under the mesh resolution in the wave tank, the mesh is only refined around the breakwater in the medial mesh group and the fine mesh group. In the medial mesh group, the meshes were refined once around the breakwater boundary. In addition, for the fine mesh group, the mesh near the breakwater boundary was refined as twice the minimum grid size of 0.005 m. The simulations are taken in parallel, competing with 4 cores on the platform of the personal computer. The detailed set-up can be seen in Table 2, and the result is shown in Fig. 5. Taking the heave motion response as the criterion, the solution achieves convergence, except for a slight phase shift that is observed in the coarse mesh group. Therefore, the next mesh resolution was set as the fine mesh group.

3.2.2 Verification case

In the verification case, all the wave conditions were calculated, as listed in Table 1. The simulation results are compared with both the experimental results from Isaacson (1998) and from Ning et al. (2016), and the numerical results under the potential theory with and without viscous correction are compared with the current simulation. The heave response is

Table 2 Mesh convergence cases set-up of the mesh number, time step and time consumption

Case set	Mesh number	Time step (s)	Time consumption (s)
Coarse mesh	27,825	Adjust time step	623
Medium mesh	38,547	Adjust time step	1695
Fine mesh	44,361	Adjust time step	4668

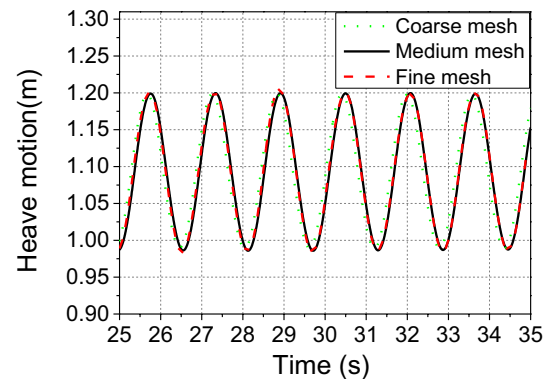


Fig. 5 Comparison of the heave motion with time using three different mesh densities

shown in Fig. 6, and the transmission coefficient is shown in Fig. 7. As is shown in Fig. 6, the result is in good agreement with the experimental data from Isaacson and from Ning. The RMSE is 0.483 with Isaacson's experimental data and is 0.1222 with Ning's experimental data. It can be seen that the floating breakwater does not achieve an apparent resonance motion. The motion achieves the same amplitude as the wave, as the wavelength increases. Defining RAO is the ratio of the response motion and the wave height ξ/H . The RAO gets close to 1. Furthermore, the numerical results without viscous correction have a large difference with the experimental data and the current simulation under viscous flow. In the zone near the resonant wave period, the result is far away from the experimental data. The numerical result with 10% viscous correction has a better result compared to that without viscous correction. A good agreement can be found in the zone, where B/L is smaller than 0.1 or is larger than 0.25.

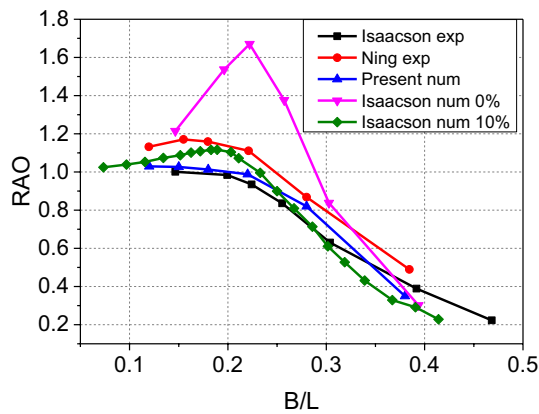


Fig. 6 Comparison of the heave RAO with B/L using present numerical model, Ning's experiment result (Ning et al. 2016), Isaacson's experiment and numerical result (Isaacson et al. 1998) using the potential with viscous correction

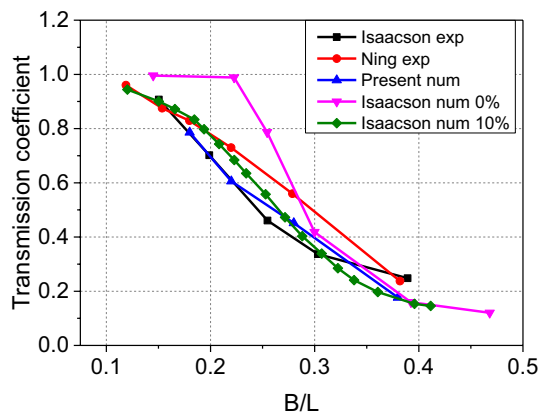


Fig. 7 Comparison of the transmission coefficient K_t with B/L using present numerical model, Ning's experiment result (Ning et al. 2016), Isaacson's experiment and numerical result (Isaacson et al. 1998) using the potential with viscous correction

Between them, there is still a slight difference. However, the result for both the motion response and the transmission coefficient under viscous flow in the paper has a very close value to the experimental data in the zone near the resonant period. Because the friction from the pile is neglected in the present simulation, a slightly larger value of the motion response is expected than that of the experimental data from Isaacson. Therefore, the model using the CFD method is verified in this paper. The result also indicates that the viscous and flow separation have a large influence on the motion of the floating

breakwater, especially near the resonant wave period. Hence, in this paper, to study the integrated system of the OBC WEC with the floating breakwater, we focus on the zone where the structure may achieve resonance, and the CFD simulation is taken to research the wave energy capture efficiency and the wave damping function of such an integrated system.

4 Results and Discussion

Considering the model scale, the scale overestimates the effect of the viscous force. Hence, in this part, following the Froude scaling, the model is amplified 10 times to become a prototype integration system that can further reduce the scale effects. The wave conditions are also amplified with the same scale. The dimensions of the prototype integration system are a length of 8.0 m, a height of 6.0 m, and a draft of 2.5 m. The numerical wave tank is 250 m length, with a water depth of 10 m. According to Isaacson's work, the floating breakwater is often operated at the place where the common wave period is 4–5 s and where the wave height is 1.0 m (Isaacson et al. 1998). Therefore, the wave conditions in the present work are shown in Table 3, where the wave period varies from 4.03 s to 6.32 s, the non-dimensional number kh varies from 1.178–2.513, and the wave height is 1.0 m.

After the tests for the accuracy of the wave generation, the mesh convergence tests and the selection of a turbulence model, as the Re number is scaled more than 50 times up to 4×10^6 compared with that in the laboratory test, the SST $k - \omega$ turbulence model is chosen in the prototype simulation. The standard wall function is applied on the boundary of the floating breakwater. The background mesh sizes are set to uniform $0.4m \times 0.4m$. The mesh is refined twice near the water-free surface and five times around the boundary of the structure as shown in Fig. 8 to limit the y^+ value between 30 and 300 (Launder and Spalding 1983). The boundary layer mesh sizes are closed to 0.0125 m in both x and z directions. The effect of the turbulence will be discussed later in subsection 4.4. The time step adjusts based on the maximum Courant number of 0.5 and on the setting of the maximum time step to 0.025 s. The PIMPLE algorithm is applied to solve the pressure field and the velocity field. The outer correction loop is set to 1, and the inner correction loop is set to 3.

Hence, all the above set-ups are approved to be used in the research work. The PTO effect on the integrated system will be studied in detail in next subsections.

Table 3 Wave conditions set-up for the prototype model

d (m)	T (s)	Kh	H_i (m)
2.5	4.032, 4.332, 4.569, 4.705, 4.850, 4.996, 5.227, 5.66, 6.325	2.513–1.178	1.0

Fig. 8 Mesh configuration of the prototype model and close-up view

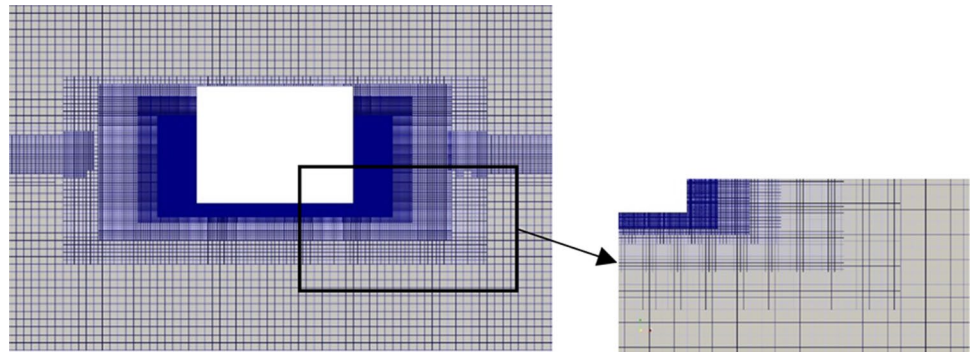


Table 4 Optimal PTO coefficient in potential flow from $kh = 1.18$ – 2.51

Kh	T	λ_{optimal}
1.18	6.325	47,888.00
1.41	5.660	33,767.92
1.60	5.227	25,799.78
1.73	4.996	21,813.24
1.81	4.850	20,144.66
1.90	4.705	18,568.66
2.00	4.569	17,885.22
2.20	4.332	18,934.62
2.51	4.032	24,617.94

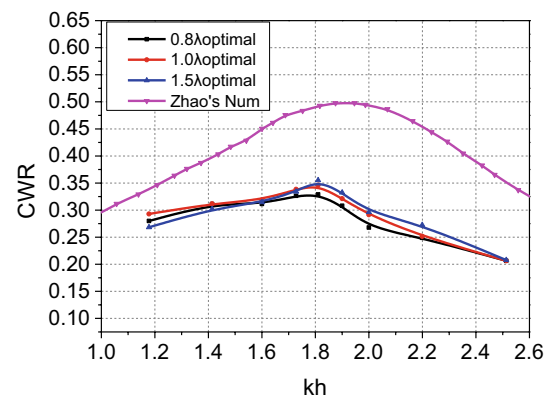


Fig. 9 Evolution of CWR with different times of optimal PTO coefficients ($0.8\lambda_{\text{optimal}}$, $1.0\lambda_{\text{optimal}}$, $1.5\lambda_{\text{optimal}}$) with Zhao's numerical one based on potential theory

4.1 The effect of the optimal PTO damping force

In this section, the effect of the optimal PTO damping force on the energy capture and the transmission coefficient is given a detailed description. Based on the introduction in Sect. 2, there is the theoretical optimal PTO damping force λ_{optimal} to the integrated system of WEC and floating breakwater. With the optimal PTO damping force, the energy capture efficiency can reach the maximum. Therefore, when designing the WEC device, the best case is that the PTO system can turn to the optimal PTO damping coefficient, corresponding to transient wave conditions. The optimal PTO damping coefficient can be derived from the frequency solution by potential theory which is listed in Table 4, from $kh = 1.18$ to 2.51 . This range of kh is where the structure may resonate under the action of waves. However, the theoretical optimal PTO damping coefficient may not be optimal under the viscous effect and flow separation. The damping needs to further explore the optimal PTO damping coefficient under the effect of viscous fluid and flow separation. Therefore, in this section, different times of the theoretical optimal PTO damping coefficient are applied to the integrated system to study the motion response, the energy capture efficiency, and the transmission coefficient.

Figure 9 is the present CWR value for the cases of $0.8\lambda_{\text{optimal}}$ and $1.5\lambda_{\text{optimal}}$ compared with the theoretical CWR value under potential theory. From the case of $1.0\lambda_{\text{optimal}}$, it

can be inferred that the CWR value has a significant decline compared to the theoretical value. The theoretical maximum CWR value reached up to 0.5 when the floating breakwater achieved resonance. However, under the viscous effect, the maximum CWR value is only 0.36, which is nearly 20% under the theoretical value. Especially near the resonance zone when kh is close to 1.81, the viscous dissipation becomes very intense, and a larger difference appears between the result under viscous flow and that with potential theory. For the cases of $0.8\lambda_{\text{optimal}}$ and $1.5\lambda_{\text{optimal}}$, the results show that a larger CWR value can be obtained under $1.5\lambda_{\text{optimal}}$ when $kh \geq 1.81$. This result means the optimal PTO coefficient changes under the viscous effect and that the optimal PTO coefficient should be larger to obtain high wave energy capture performance (Isaacson et al. 1998). However, when $kh < 1.8$, the CWR value under $1.5\lambda_{\text{optimal}}$ is smaller than is the case under $1.0\lambda_{\text{optimal}}$. This result indicates that the integration system is over-restrained by the PTO system with $1.5\lambda_{\text{optimal}}$. Hence, the CWR declines when $kh < 1.81$. The result also means that the optimal PTO coefficient for a range of wave conditions cannot just be a simple amplification of the optimal PTO coefficient under potential theory. Further studies focus on the exact optimal PTO coefficient

at $kh = 1.81$. Figure 10 shows the CWR value under several times of λ_{optimal} . It can be clearly seen that the maximum CWR value occurs at $1.5\lambda_{\text{optimal}}$ and begins to decrease at $2.0\lambda_{\text{optimal}}$. The result also shows that the CWR is not very sensitive in the range of $1.0\lambda_{\text{optimal}} - 2.0\lambda_{\text{optimal}}$, but it is very sensitive at $0.8\lambda_{\text{optimal}}$. Hence, the $1.0\lambda_{\text{optimal}}$ value can be chosen as the minimum value, and a larger PTO value than the $1.0\lambda_{\text{optimal}}$ is allowed when considering the viscous and flow separation effects. Figure 11 shows the transmission coefficient for the cases of several times of the PTO coefficient. The transmission coefficient declines from 0.55 to 0.45 at $kh = 1.81$ for the case of $1.0\lambda_{\text{optimal}}$. Further, when the PTO coefficient increases to $1.5\lambda_{\text{optimal}}$, it decreases to 0.4. Unlike the potential theory, whose transmission coefficient is an outward curve, a novel phenomenon is that the transmission coefficient is an inward curve near the zone of $kh = 1.81$. The curve is more obvious when the PTO coefficient increases, as in the case of $1.5\lambda_{\text{optimal}}$. This outcome may be caused by the increases in wave energy dissipation at this zone. Thus, a proper increase in the PTO coefficient can dramatically decrease the transmission coefficient and can increase the useful range of the device under a wider range of wave conditions. If the criterion for the transmission is set to under 0.5, the useful range of the wave conditions increases from $kh > 1.9$ to $kh > 1.6$, which has a profound benefit to the operation of the device in a real sea state.

4.2 Uniform PTO Damping Coefficient

For most WEC devices, it is not easy to transform the PTO damping coefficient into the optimal value corresponding to the instantaneous wave conditions. In contrast, the coefficient is often a fixed value during the operation life. Hence, to study the wave energy capture efficiency and the wave dissipation under a fixed PTO damping coefficient, the PTO damping coefficient $1.5\lambda_{\text{optimal}}$ at $kh = 1.81$ is applied in this

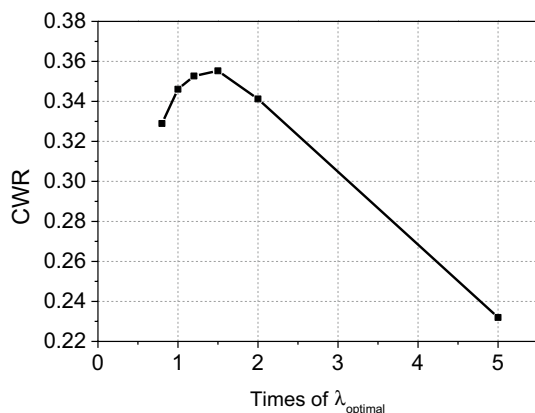


Fig. 10 Evolution of the maximum CWR value with different times of optimal PTO damping force at $kh = 1.81$

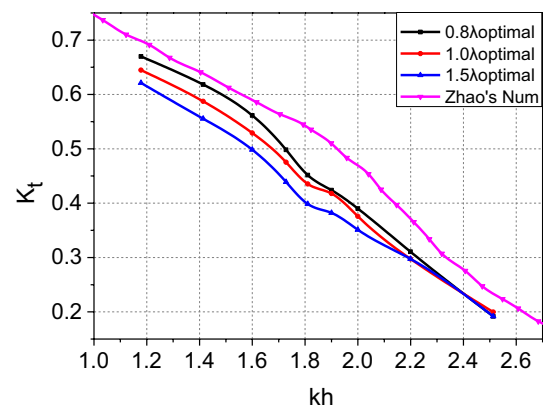


Fig. 11 Evolution of transmission coefficient K_t with different times of optimal PTO coefficients ($0.8\lambda_{\text{optimal}}$, $1.0\lambda_{\text{optimal}}$, $1.5\lambda_{\text{optimal}}$) with Zhao's numerical one based on potential theory

section as the fixed PTO damping coefficient, as it is the PTO coefficient that achieves a maximum CWR value in the above simulation case.

The CWR value for the fixed PTO case compared with the case under $1.5\lambda_{\text{optimal}}$ is shown in Fig. 12. When $kh > 1.81$, the value is a short-wavelength zone. The result for the CWR value is almost equal for the two cases. This result maybe because the PTO damping coefficient has a small difference for the short-wavelength zone, as shown in Table 4, when $kh = 1.81 - 2.51$. However, for the cases where $kh < 1.81$, there is an apparent difference between the two cases. A higher CWR value can be obtained in the fixed PTO damping coefficient case compared with the case under $1.5\lambda_{\text{optimal}}$. As the fixed PTO coefficient value is smaller than the value of $1.5\lambda_{\text{optimal}}$ when $kh < 1.81$, this also means that the WEC device is over-restrained in the long-wavelength zone, which corresponds to the decrease in the CWR value for the $1.5\lambda_{\text{optimal}}$ case in the above section. The result also indicates

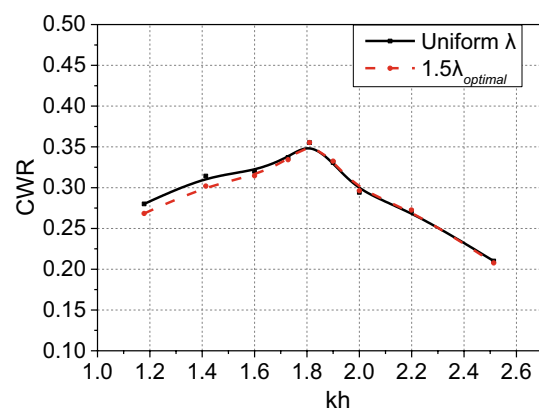


Fig. 12 Comparison of the CWR value with kh between two cases (uniform λ and $1.5\lambda_{\text{optimal}}$)

that the transformation of the optimal PTO damping coefficient from the potential theory to the viscous flow is also nonlinear. The system needs to consider the optimal PTO value for the long-wavelength zone and the short-wavelength zone. From the result of the CWR value, it appears that the fixed PTO system has better performance. However, it can be shown from Fig. 13 that the motion response can dramatically increase at the long-wavelength zone when $kh < 1.81$. The large motion response may lead to a larger transmission coefficient, as shown in Fig. 14; additionally, the large motion responses may be unfavourable factors for the stability and durability of the device, as the CWR value has only a slight difference between the fixed PTO coefficient case and the optimal PTO coefficient case. However, the increase in the motion response and the transmission coefficient is significant. This outcome suggests that adjusting the PTO coefficient to an optimal value is still necessary to optimize the performance of the integration system.

4.3 Nonlinear PTO System

The Coulomb-type PTO damping model is developed and verified in this section. The wave condition $T = 4.85$ s is selected for the verification case. F_{PTO} for the Coulomb PTO model is set as $\lambda_{optimal} \times V_{peak}$. $\lambda_{optimal} = 20187.5$ Ns/m is the optimal linear PTO coefficient in the case of $T = 4.85$ s. $V_{peak} = 0.4$ m/s is the peak velocity of the floating breakwater in the simulation of wave condition $T = 4.85$ s. Figure 15 shows the velocity and PTO damping force from 70 s to 110 s. The result shows that the Coulomb damping system can be numerically stable in the current simulation. A convergent result can be obtained for both the velocity and the PTO damping force. The phase of the PTO damping force is also synchronous with the velocity. To study the difference effect of the Coulomb PTO model and the linear PTO model on the performance of the integration system, Fig. 16 shows

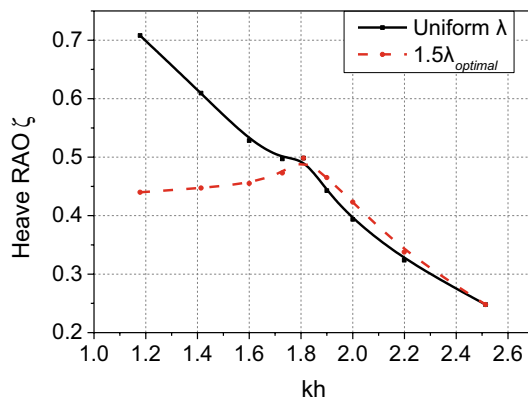


Fig. 13 Comparison of the heave RAO value with kh between two cases (uniform λ and $1.5\lambda_{optimal}$)

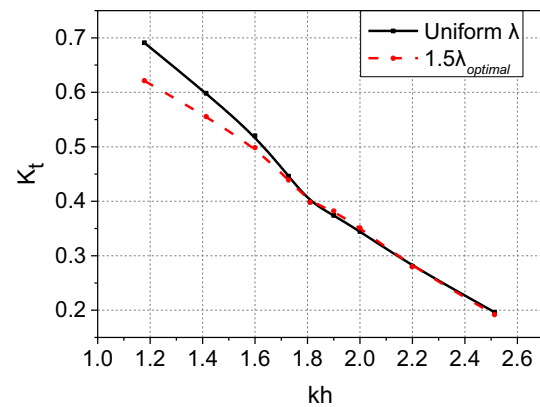


Fig. 14 Comparison of the transmission coefficient K_t between two cases (uniform λ and $1.5\lambda_{optimal}$)

the comparison of the velocity of the floating breakwater for both the linear PTO model and the Coulomb PTO model. The motion response under the Coulomb damping system is smaller than the linear one. In addition, the motion velocity curve shows a slightly nonlinear pattern, unlike the linear one that is a sinusoidal waveform. This result is because the PTO force of the Coulomb model remains constant in the half period of the motion. In addition, the force remains a maximum value in the linear model. Therefore, the motion response will be smaller, by approximately 20%, than the linear response. A smaller motion response with a larger CWR value, equal to 0.349, can be obtained from the integration system with the Coulomb damping model compared with the linear one, in which the CWR value is 0.346. This result shows that the Coulomb damping model can also have as good of a performance as the linear PTO model. The smaller motion of the integration system under the Coulomb damping force can lead to a lower wave transmission coefficient. The wave transmission coefficient decreases from

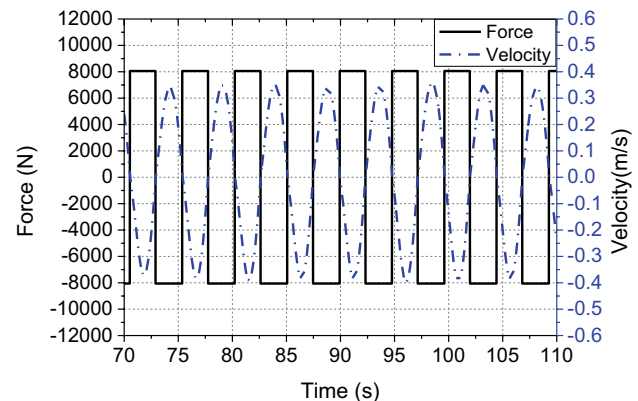


Fig. 15 Time history of the PTO Coulomb damping force and the buoy moving velocity

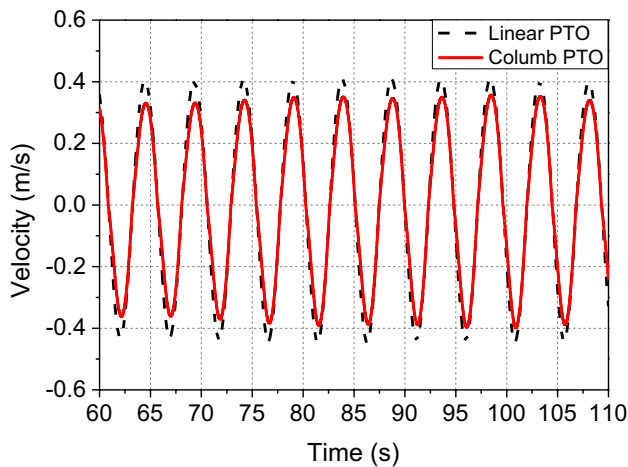


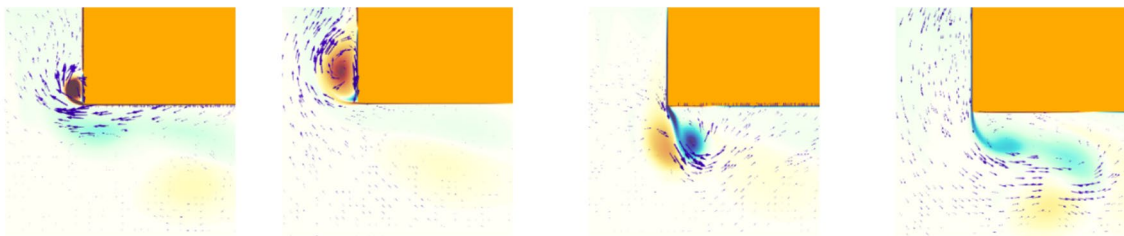
Fig. 16 Comparison of the moving velocity from the case with Linear PTO damping force and the one with Coulomb-type PTO damping force

0.435 to 0.39, by nearly 10%. The hydraulic PTO damping model can easily implement the auto-phase control strategy to achieve a higher CWR. Therefore, the simulation of the Coulomb PTO model, which can be a simplification of the hydraulic PTO damping model, can provide a reference to evaluate the performance of such an integration system in a real sea state.

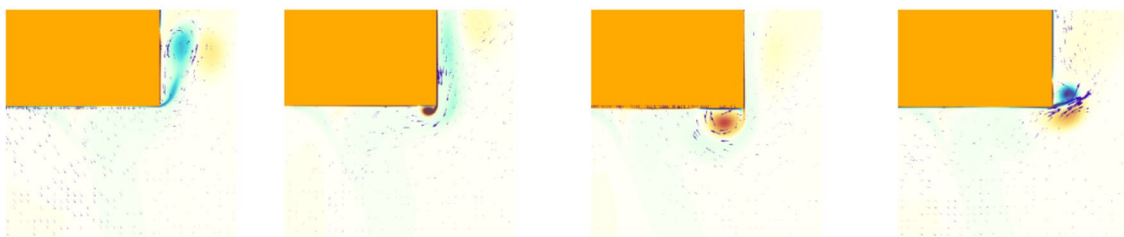
4.4 Turbulence effects

The prototype model which is scaled 10 times from the laboratory model is simulated in this section. The turbulence model is applied to permit a more accurate solution of the flow field. The effect of the turbulence is studied in this subsection. The eight typical vorticity fields near the front and the rear corner of the buoy during a heave movement cycle are shown in Fig. 17a, b. It shows strong vortex generation in the front corner of the buoy as well as the back corner. Due to the wave field near the bottom of the buoy, the vortex generation has phase difference unlike the heave oscillating buoy in the station water where the vortex generation will be symmetric. Because of the strong vorticity at the corners, the energy dissipation can be enhanced by the turbulence flow as shown in Fig. 18. The CWR value of the prototype model using the turbulence model is compared with the laboratory model with laminar model in Fig. 19. It shows that the scale effect is not significant in the scale of 1:10, but the turbulence model produces extra energy dissipation in the prototype model, especially in the short wave conditions when taking a long time simulation. This may be due to the shortage of the SST $k - \omega$ model in the free surface flow which is over-produced turbulence as reported in Larsen's work (2018) about the analysis of the over-production of turbulence beneath surface waves. The new turbulence model can be

(a) Front corner



(b) Rear corner



86.28 s

87.5 s

88.71 s

89.93 s



Fig. 17 The snapshot of the vorticity field and the velocity vector near the bottom of the buoy at four specific moments of a motion cycle at wave condition $T=4.85$ s, $H=1.0$ m when the buoy achieves resonance

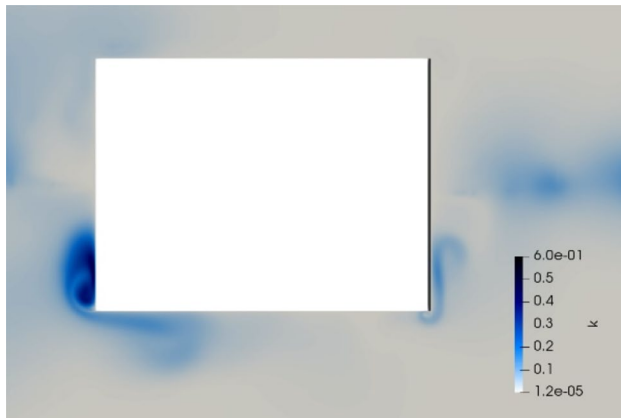


Fig. 18 Turbulent kinetic energy field near the bottom of the buoy at time 87.5 s

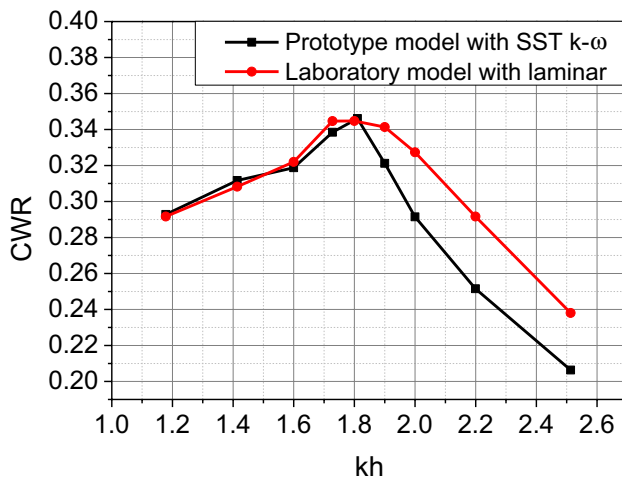


Fig. 19 Comparison of the CWR value with kh between the numerical model and the laminar laboratory model

imposed in present work in the future work to accurately perform the simulation.

5 Conclusions

The hydrodynamic performance of the integration system that includes an oscillating buoy wave energy converter with a pile-restrained floating breakwater is simulated using OpenFOAM in the present work. The effect of the PTO system is systematically studied by, respectively, considering the optimal PTO coefficient under viscous flow, under fixed PTO coefficient over wave conditions, and under the Coulomb damping PTO model. The results can be described as follows:

- (1) When considering the viscous effect, the CWR will achieve a significant decrease compared to the theoretical optimal CWR under potential flow. Thus, it is crucial to consider the viscous effect to evaluate the energy capture efficiency of the WEC device.
- (2) Because of the viscous effect, the linear optimal PTO coefficient should be larger than the theoretical optimal coefficient in potential flow.
- (3) The fixed PTO cases can also achieve a similar CWR value to the optimal PTO cases. This result means that the CWR value is not very sensitive to the PTO coefficient. For some designs of a single WEC device, it is not necessary to change the PTO coefficient corresponding to the instant wave condition. However, for the integration system, when choosing the fixed PTO system, larger motion responses and lower transmission coefficients will have disadvantages to coastal protection.
- (4) The development work of the Coulomb damping model in OpenFOAM enriches the function of OpenFOAM for modelling WEC devices. The stable value of the velocity and the PTO force from the simulation can be regarded as a verification of the Coulomb damping model in OpenFOAM. A rough comparison of the Coulomb damping model and the linear damping model shows that the Coulomb damping model has a good performance in terms of both wave energy capture efficiency and coastal protection.

The effect of the Coulomb damping model on the results is complicated due to coupled dynamic between damping and motion, and then, the present model can be extended to further study the relationship between Coulomb damping model and hydrodynamic features. In addition, the shape of the device still needs to be optimal to minimize the viscous effect.

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